

Chapter 2: Probability Distributions and Sampling Distribution

2.1 Random Variables:

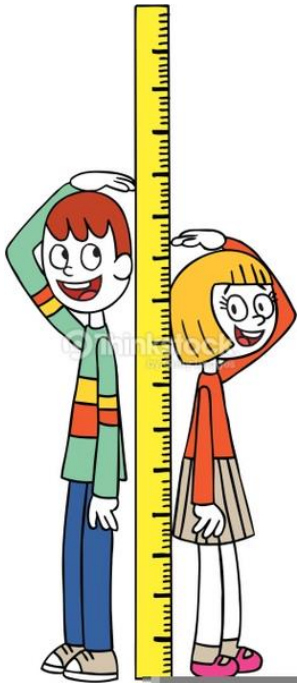
- The outcome of an experiment cannot be predicted
- The outcome of an experiment need not be a number
- Outcomes are represented as numbers.
- A random variable is a function that associates a unique numerical value with every outcome of an experiment.

- A variable whose values are determined by the outcomes of a random experiment is random variables.
- The value of the random variable will vary from trial to trial as the experiment is repeated.
- Random variables denoted by capital letters X , Y , Z
- In toss of coin, the outcomes can be denoted as Head by 1 and Tail by 0.

- Two types of random variable

Discrete and Continuous

- Discrete random variables can take only a countable number of distinct values, such as 0,1, 2,.....etc.
- Number of attendees, number of defectives, number of children in a family are discrete random variables.



Discrete Probability Distribution

- A discrete probability distribution is a list of probabilities that is associated with each outcome of an experiment.
- The list of probabilities can be shown by tabular form, graphical form or in equation form.
- It is also known as Probability Mass Function (pmf).
- Probabilities are denoted by $P(X = x)$, where X is random variable and x is its possible value.

Properties of Discrete Probability Distribution

- $\sum P(X=x) = 1$ [sum of all probabilities of possible events]
- $0 \leq P(X=x) \leq 1$ [probability lies between 0 and 1]

Examples

1) The missing probability value will be

RV X	1	2	3	4
Probability, $P(X)$	0.25	0.50	0.15	?

2) In tossing three coins, the probability distribution of getting heads

RV X (No. of heads)	0	1	2	3
Probability, $P(X)$	$1/8$	$3/8$	$3/8$	$1/8$

Cumulative Distribution Function

- Evaluation $P(X \leq x)$ yields cumulative distribution function because we add probabilities of all X which are at most x .
- $F(x) = P(X \leq x)$ = Probability that the observed value of X will be at most x
- In three tosses of coin probability of getting at most 2 heads is $7/8$

2.2 Expectation and variance of discrete random variables

- If X is a discrete random variable, expected value is the average value that a r.v. can take.
- It the average value of probability distribution
- Denoted by $E(X)$
- $E(X) = \sum x \cdot P(X_i)$
- Similar as $\bar{y} = \sum fx/n$
- Expected number of heads (average number of toss) in three tosses of coin = $3/2 = 1.5$

Variance

- It describes the variability of the probability distribution.
- Denoted by $V(X)$
- $V(X) = E(X^2) - [E(X)]^2$
Where $E(X^2) = \sum X^2 P(X=x)$
- Similar to σ^2

Properties of Expectation

- If X and Y are random variables, then $E(X + Y) = E(X) + E(Y)$
- If X and Y are independent random variables, then $E(XY) = E(X).E(Y)$
- If X is a random variable and a and b are constants, then $E(aX + b) = a E(X) + b$
- $E(a) = a$
- $V(aX \pm c) = a^2 V(X)$, where c is constant
- $V(aX) = a^2 V(X)$, where a is constant
- $V(c) = 0$, where c is constant

Exercise

1. A random variable X has the following probability mass function:

$X = x$	1	2	3	4	5	6	7
$P(X=x)$	a	$2a$	$2a$	$3a$	a^2	$2a^2$	$7a^2+a$

Find (i) the value of a (ii) mean and variance of X (iii) $P(X \geq 2)$, $P(X \leq 3)$, $P(X < 3)$

2. By investing in a particular stock, a person can make a profit in one year of Rs 4000 with probability 0.3 or take a loss of Rs 1000 with probability 0.7. What is the person's expected gain?
3. From a lot of 10 items containing 3 defectives, a sample of 4 items is drawn at random. Let the random variable X denote the no. of defective items in the sample. Answer the following when the sample is drawn (without replacement)
- Find the probability distribution of X
 - Find $P(X \leq 1)$, $P(X < 1)$ and $P(0 < X < 2)$
 - Obtain the mean and variance for the distribution.

Continuous Random Variables and Probability Distributions

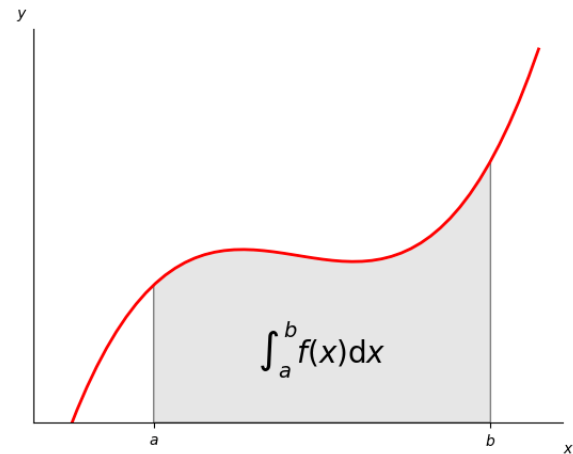
A random variable X is said to be continuous

- If it's set of possible values is an entire interval of numbers
- If it's possible values cannot be put into one-to-one correspondence.
- Variable whose value which can be measured
- For example, height, weight, length, time, intensity of light etc.

Probability Density Function (PDF)

- If X is a continuous random variable, then prob. Density function for any two numbers a and b ($a < b$) is given by

- $$P(a \leq X \leq b) = \int_a^b f(x) dx$$



- $y = f(x)$ is called probability density curve
- $P(a \leq X \leq b)$ gives the area under the curve that lies between the ordinate $x = a$ and $x = b$.

Note:

- $f(x) \geq 0$

- $\int_{-\infty}^{+\infty} f(x)dx = 1$

- Probability at a point is zero, because

$$P(X=c) = \int_c^c f(x)dx = 0$$

Cumulative Distribution Function (CDF):

- Evaluation $P(X \leq x)$ yields cumulative distribution function because we add probabilities of all X which are at most x .
- $F(x) = P(X \leq x)$ = Probability that the observed value of X will be at most x

- $$F(x) = P(X \leq x) = \int_{-\infty}^x f(x) dx$$

Properties:

- $0 \leq F(x) \leq 1$
- $d/dx F(x) = f(x)$
- $P(a \leq x \leq b) = P(a < x \leq b) = P(a \leq x < b) = P(a < x < b) = F(b) - F(a)$
- $F(-\infty) = 0, F(+\infty) = 1$

Expected Values

- If X is a continuous random variable, expected value is the average value that a r.v. can take.

- $$E(X) = \int_{-\infty}^{+\infty} x f(x) dx$$

- $$\text{Var}(X) = E(X^2) - [E(X)]^2$$

Where
$$E(X^2) = \int_{-\infty}^{+\infty} x^2 f(x) dx$$

2.3 Discrete probability distributions

Binomial Probability Distribution

Characteristics:

- Each trial results in two mutually exclusive cases, termed as success and failure.
- Probability of success “ p ” is constant for each trial.
- Trials are independent of each other.
- Number of trials is finite.

Let n trials be performed and the occurrence of an event in a trial be called success and its non-occurrence be failure.

p = Probability of success in any trial

q = Probability of failure in any trial = $1 - p$

The Probability of x success and $n-x$ failures in some specific order is given by

$$\begin{aligned} P(\text{SSF} \dots \text{SF}) &= P(S) \cdot P(S) \cdot P(F) \dots P(S) \cdot P(F) \\ &= p \cdot p \cdot q \dots p \cdot q \\ &= \underbrace{p \cdot p \cdot p \dots p}_{(x \text{ factors})} \underbrace{q \cdot q \dots q}_{(n-x \text{ factors})} \\ &= p^x q^{n-x} \end{aligned}$$

- But there are $C(n, x)$ ways in which x success in n trials can occur. Hence, the probability of x success in n trials in any order is given by $C(n, x) p^x q^{n-x}$

$$P(X = x) = C(n, x) p^x q^{n-x} ; x = 0, 1, 2, \dots, n \\ = 0, \text{ otherwise}$$

Mean & Variance of Binomial Distribution:

- Mean = $E(X) = np$
- Variance = $V(X) = npq$

Negative Binomial Distribution

Characteristics

- Each trial can result in either a success (S) or a failure (F).
- Trials are independent of each other.
- Probability of success is constant from trial to trial.
- The experiment continues until a total of “ r ” success has been observed

- The random variable x = number of failures which precede the r^{th} success in $(x+r)$ trials
- In this distribution, number of success is fixed and number trials are random
- Probability of x failures before r^{th} success in any order is

$$P(X=x) = C(x+r-1, r-1) p^r \cdot q^x ; x = 0,1,2,3,\dots$$

Mean & Variance of Negative Binomial Distribution

- Mean = $r(1-p)/p$,
- Variance = $r(1-p)/p^2$

Question

- What are the differences and similarities between Binomial and Negative Binomial Distribution?

OR

- Compare between Binomial and Negative Binomial Distribution.

Poisson Probability Distribution

Chief Characteristics

- Events take occurrences over a specified interval. The interval can be time, distance, area, volume, or some similar unit.
- The probability of one event in the interval is proportional to the length of the interval.
- Number of outcomes occurred in those interval is independent of those occurring in other unit interval.
- The probability of success in those units is very small.

- The random variable x = number of occurrences of the event in an interval.
- A random variable X is said to have a Poisson distribution if the p.m.f. of X is given by

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!} ; x = 0, 1, 2, \dots$$

Where λ is the average number of occurrences of events in an interval = np

n = number of events, p = probability of success

Approximation to Poisson distribution

- As $n \rightarrow \infty$, $p \rightarrow 0$ and $np = \lambda$, is finite, binomial probability tends to Poisson probabilities
- Poisson is a good approximation of the binomial when $n \geq 20$, $p \leq 0.05$ and $np < 10$

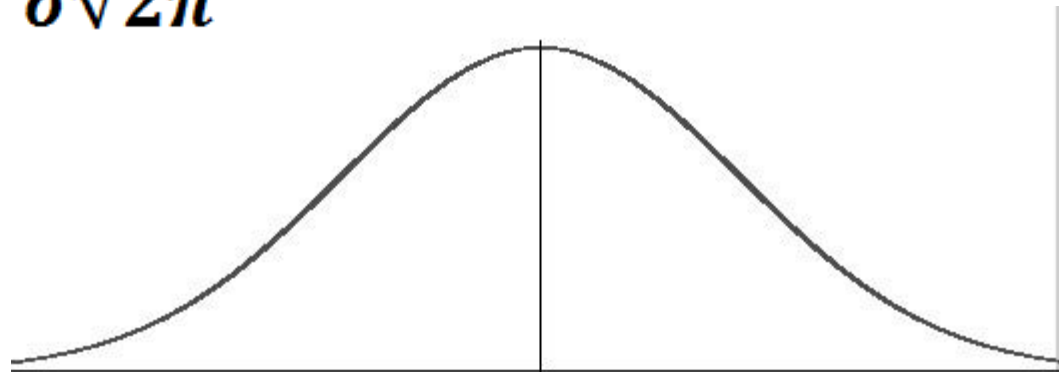
Mean and Variance: $E(X) = \text{Var}(X) = \lambda$

2.4 Continuous probability distributions:

Normal Distribution

- A random variable X having the bell-shaped distribution as shown in figure is called a Normal R.V.
- The mathematical equation for probability distribution depends upon two parameters mean (μ) and standard deviation (σ) and is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}; -\infty < x < +\infty$$

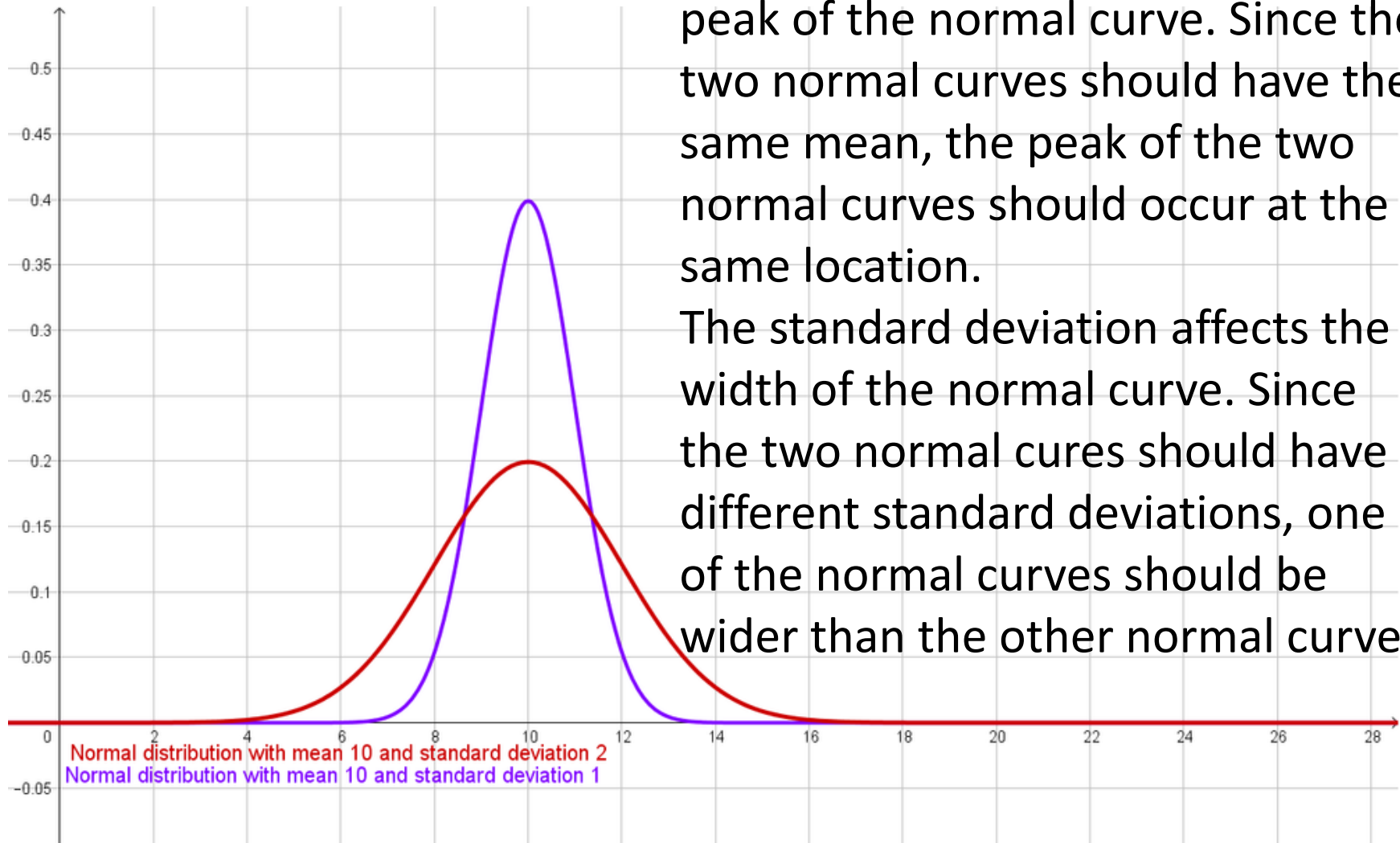


Different forms of Normal Curve

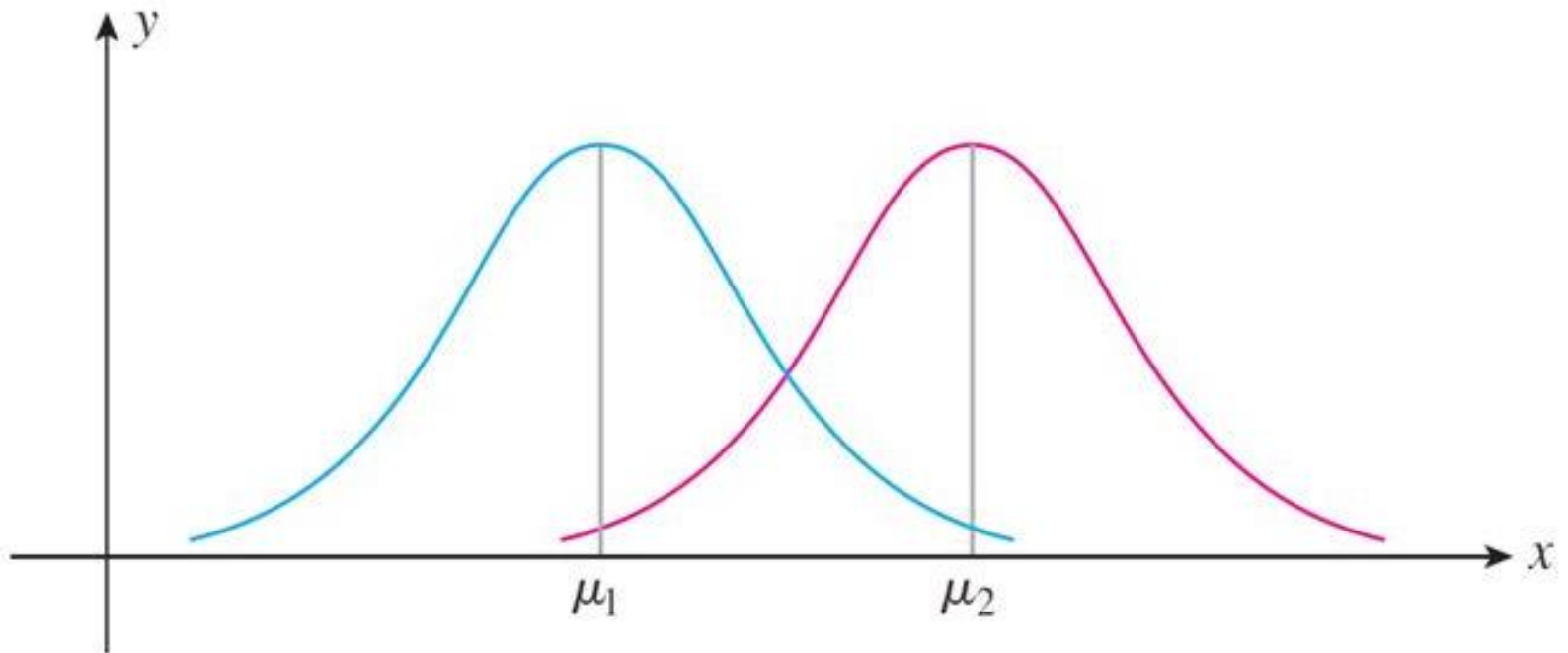
a) $\mu_1 = \mu_2$ and $\sigma_1 < \sigma_2$

The mean is represented by the peak of the normal curve. Since the two normal curves should have the same mean, the peak of the two normal curves should occur at the same location.

The standard deviation affects the width of the normal curve. Since the two normal curves should have different standard deviations, one of the normal curves should be wider than the other normal curve.

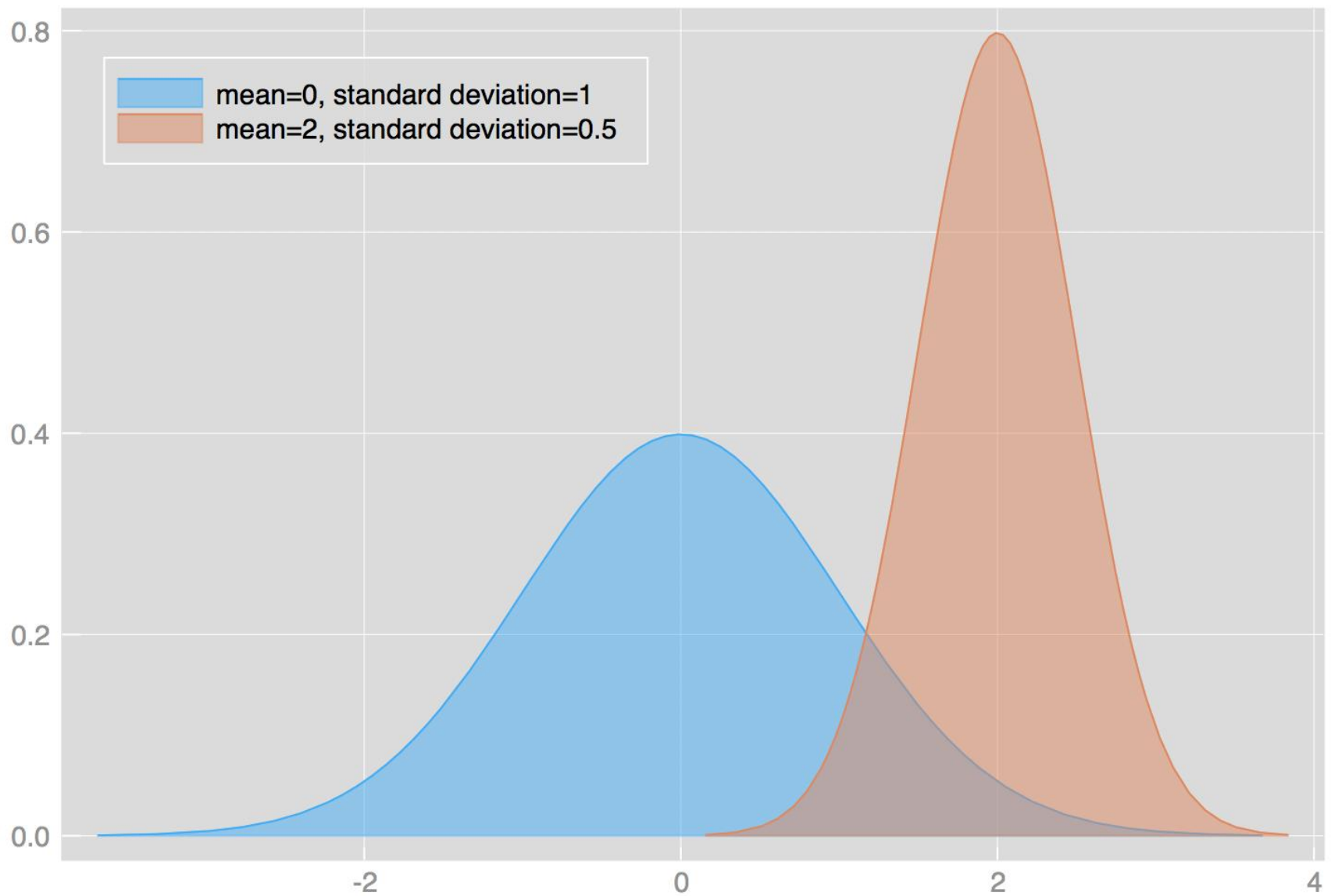


b) $\mu_1 < \mu_2$ and $\sigma_1 = \sigma_2$



Two normal curves that have the same standard deviation but different means

- $\mu_1 < \mu_2$ and $\sigma_1 < \sigma_2$



Properties of Normal Curve

- The curve is symmetrical about the line $x = \mu$
- Mean, Median and Mode of distribution coincide.
- The maximum prob. occurs at point $x = \mu$ and given by $P(X)_{\max} = \frac{1}{\sigma\sqrt{2\pi}}$
- Since $f(x)$ being prob. and can be never negative, no portion of the curve lies below the X axis.
- X-axis is an asymptote to the curve.
- Total area under the curve and above the horizontal axis is 1.

Area Property

1. About 68 % of all values lie within ± 1 standard deviation from mean.

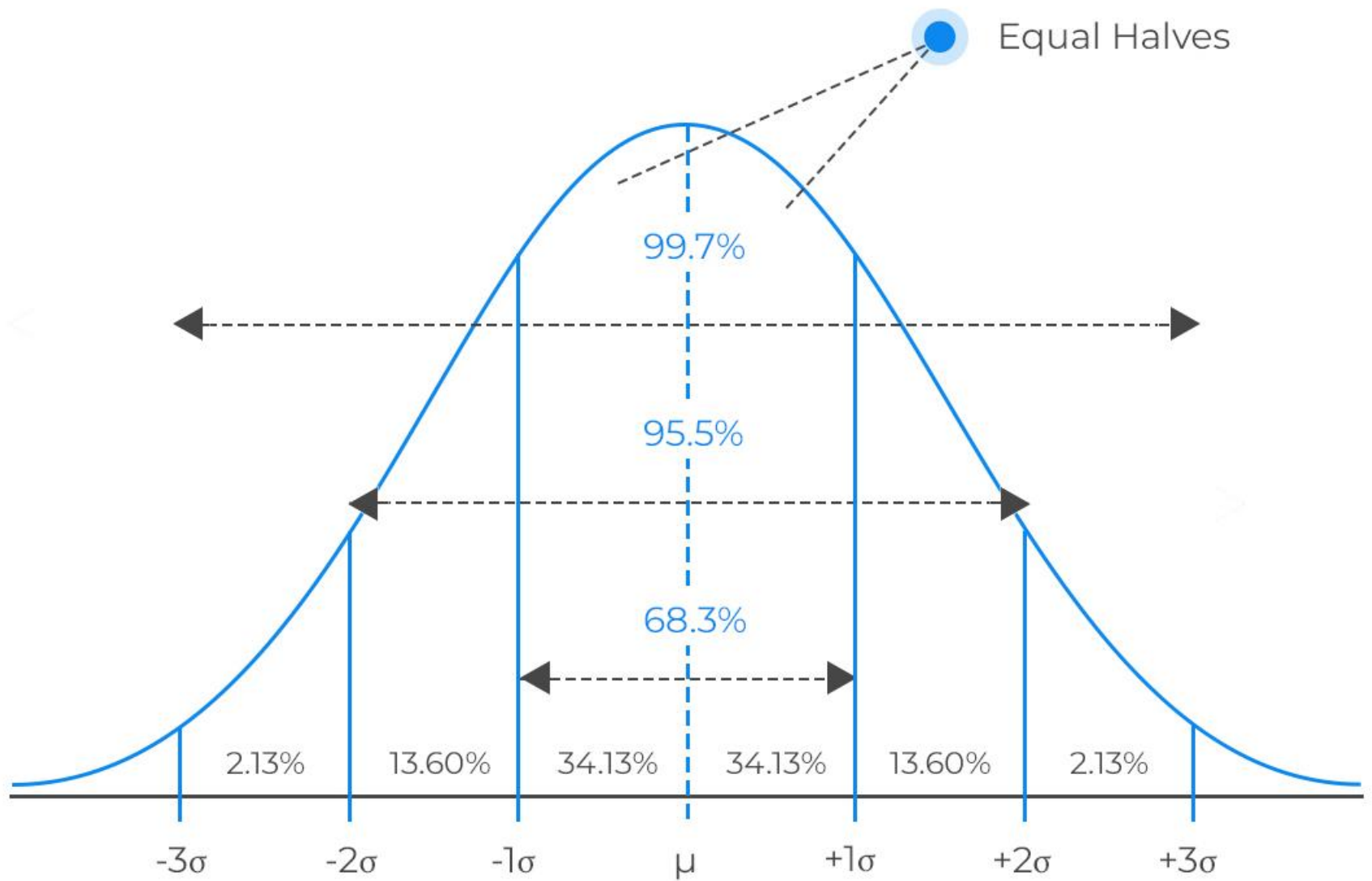
$$P(\mu - \sigma < X < \mu + \sigma) = 0.6826$$

2. About 95.5 % of all values lie within ± 2 standard deviation from mean.

$$P(\mu - 2\sigma < X < \mu + 2\sigma) = 0.9544$$

3. About 99.7 % of all values lie within ± 3 standard deviation from mean.

$$P(\mu - 3\sigma < X < \mu + 3\sigma) = 0.9973$$

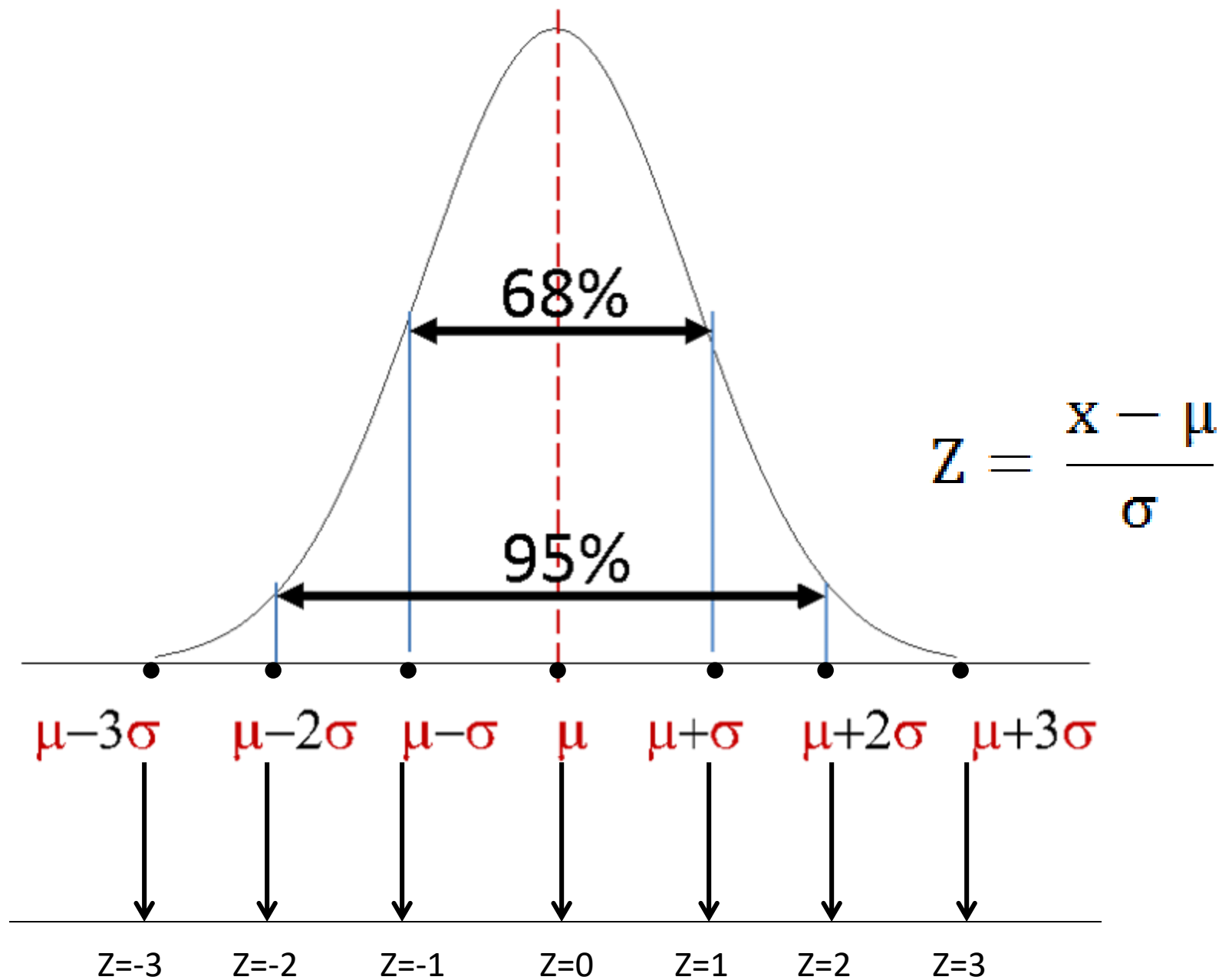


No. of standard deviations from the mean

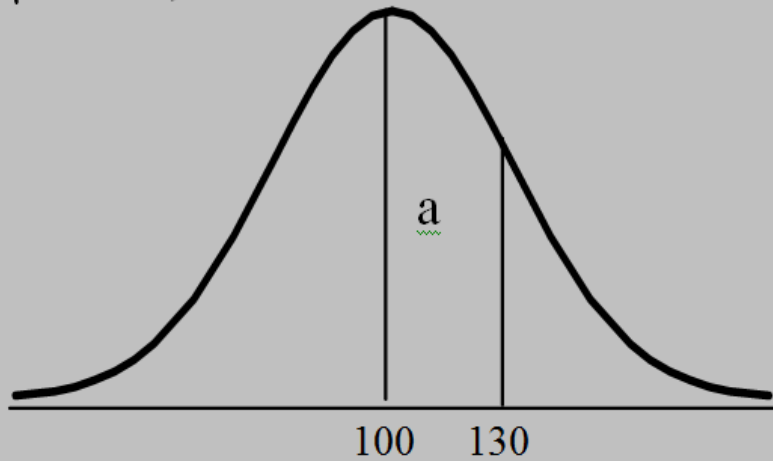
- The area property holds true for all types of normal curves.
- Whether it may be height, weight, time, temperature, voltage, if these variables follow normal distribution and have different shapes of normal curve, area property is applicable to all types of normal curves.
- This makes possible to use only one standard normal table to determine probabilities.

- This can be done by transformation of all observations of normal R.V. X to a new set of observations of normal random variable Z with mean zero and variance 1.
- This can be done by setting $Z = \frac{x - \mu}{\sigma}$
- The normal distribution with a mean of 0 and a standard deviation of 1 is called standard normal distribution.
- Normal distributions can be transformed to standard normal distributions by the formula:

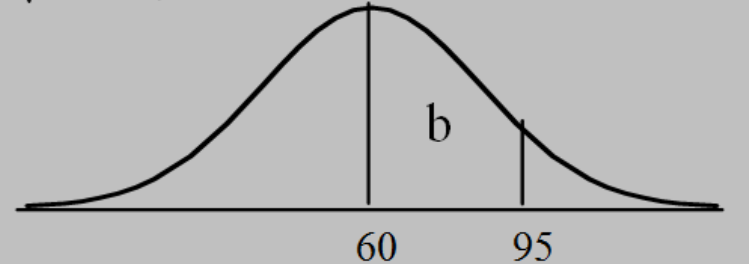
$$Z = \frac{x - \mu}{\sigma}$$



$$\mu = 100, \sigma = 30$$



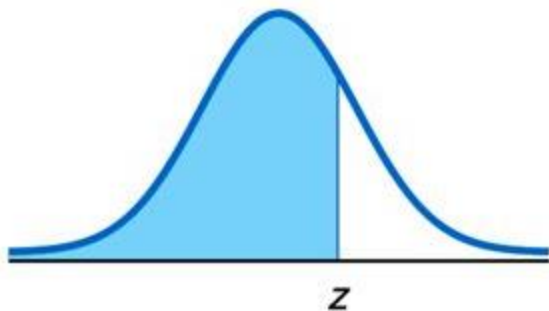
$$\mu = 60, \sigma = 35$$



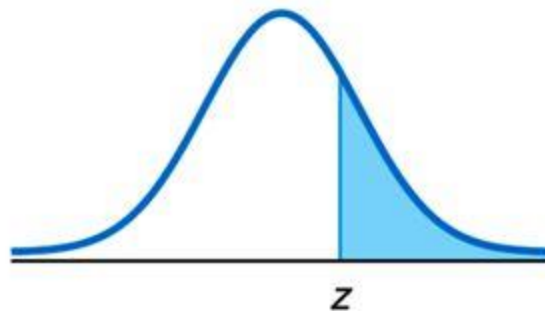
- Above figures are two normal probability distributions with different mean and standard deviation.
- Both area a & b contains same proportion of total area. Why?
- Because these are being between mean and one standard deviation to the right of μ .
- All intervals containing same number of s.d. from mean will contain same proportion of total area under curve for any normal distribution.
- This makes possible the use of only one standard normal probability distribution table.

Using Z Score Table

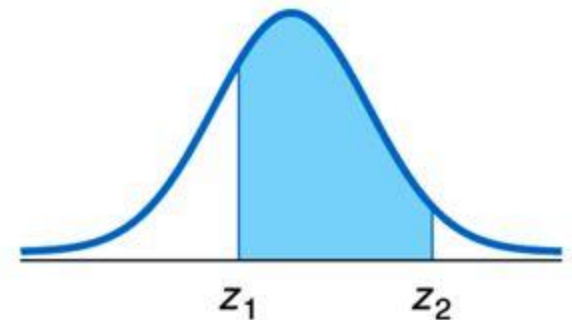
- a) $P(Z < z) =$ Shaded area left of z
- b) $P(Z > z) =$ Shaded area right of $z = 1 - P(Z < z)$
- c) $P(z_1 < Z < z_2) =$ Shaded area between of z_1 and z_2 .
 $= P(Z < z_2) - P(Z < z_1)$



(a) Shaded area:
Area to left of z



(b) Shaded area:
 $1 - (\text{Area to left of } z)$



(c) Shaded area:
(Area to left of z_2)
 $-$ (Area to left of z_1)

Approximation of Normal Distribution

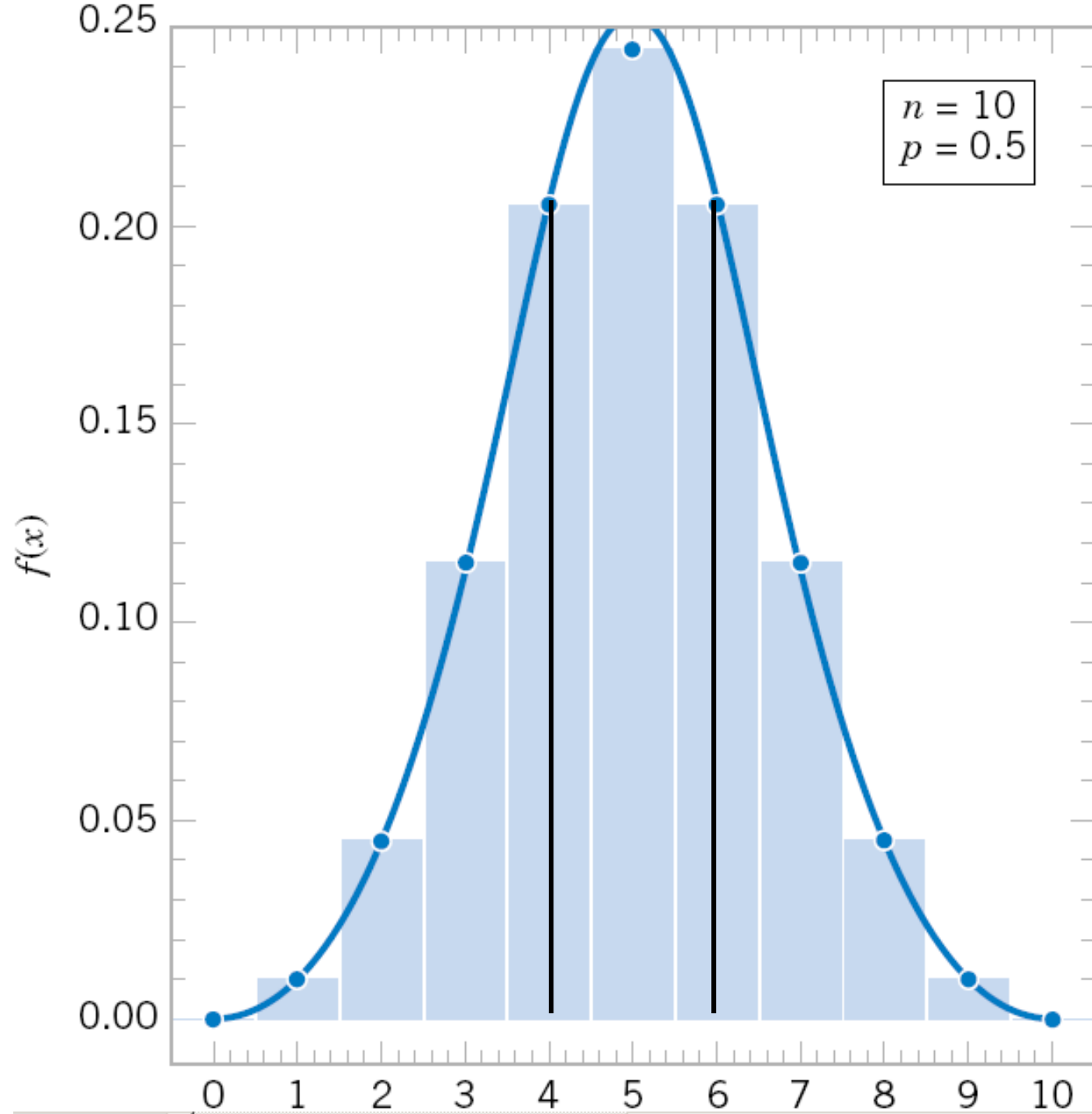
1. Approximating the Binomial Distribution

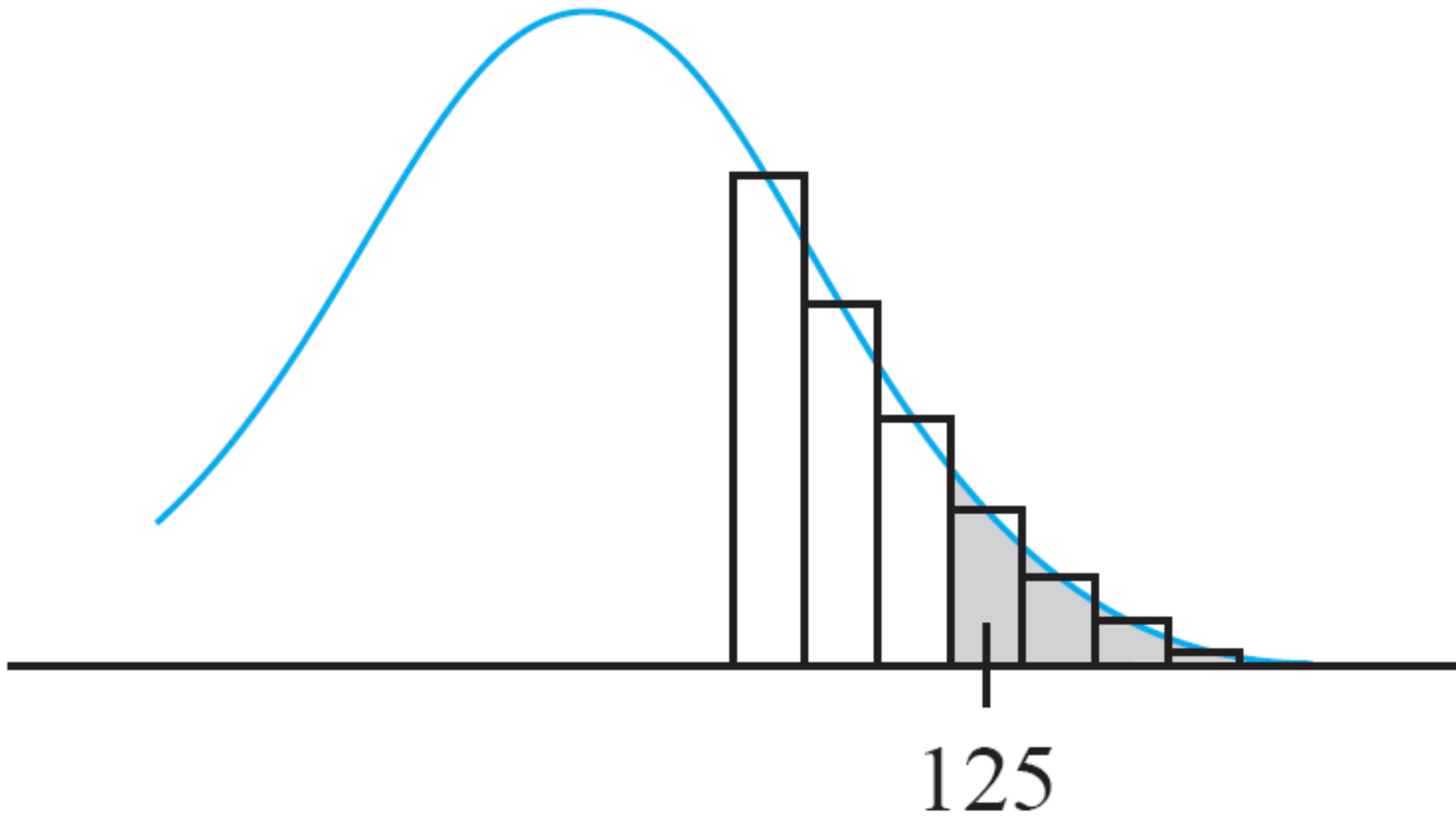
- Normal distribution can be used to approximate binomial probabilities for cases in which n is large.
- As $n \rightarrow \infty$ and neither p nor q is very small, Binomial probabilities can be approximated by Normal distribution. If X is a binomial random variable, with mean np and variance npq , then
- $Z = \frac{x - np}{\sqrt{npq}}$ is a Standard Normal variate.
- The approximation is good, if both $np \geq 10$ and $nq \geq 10$

2. Approximating the Poisson Distribution

- As $\lambda \rightarrow \infty$ Poisson probability tends to Normal distribution. If X is a Poisson random variable, with mean and variance λ , then
- $Z = \frac{X - \lambda}{\sqrt{\lambda}}$ is a standard normal variate.
- The approximation is good for $\lambda > 5$

- **Continuity Correction:** It is the correction for discreteness of distribution. If we want to approximate it's distribution with normal distribution,
- $P(X \leq x) = \text{area under the normal curve to the left} + 0.5$
- $P(X \geq x) = \text{area under the normal curve to the right} - 0.5$





The probability $P(X=125)$ would be approximated by the area between 124.5 and 125.5

Other Continuous Probability Distributions

Gamma distribution

- A continuous random variable X is said to have a gamma distribution if pdf of X is

$$f(x) = \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta}, x > 0$$

- Note: $\Gamma(\alpha) = (\alpha-1)\Gamma(\alpha-1)$
- $\Gamma(n) = (n-1)!$
- $\Gamma(1/2) = \sqrt{\pi}$
- For several pairs of (α, β) , we have a family of pdf that yields wide variety of skewed distribution called gamma family.

Mean and Variance of Gamma Distribution

- $\mu = \alpha\beta$ and $\sigma^2 = \alpha\beta^2$

- **Chi-squared distribution**

It is a special case of Gamma distribution with $\alpha = n/2$ and $\beta = 2$

$$f(x) = \frac{1}{2^{n/2} \Gamma(n/2)} x^{n/2-1} e^{-x/2}, x > 0$$

- **Exponential distribution**

It is also a special case of Gamma distribution with $\alpha = 1$

$$f(x) = 1/\beta e^{-x/\beta}, x > 0, \beta > 0$$

- **Standard Gamma Distribution**

For $\beta = 1$, we get Standard Gamma Distribution

$$f(x) = \frac{1}{\Gamma(\alpha)} x^{\alpha-1} e^{-x}, x > 0$$

2.5 Population and Sample:

- Population is a collection (or set) of data that describes some phenomenon of interest to us.
- Statistical population does not refer to people as in our everyday usage of the term; it refers to a collection of data.
- A population is the complete collection of all elements (score, people, measurements, and so on) to be studied.
- A sample is a subset of data selected from a population.

- **Parameter**: A numerical descriptive measure of a population is called parameter. Mean (μ), variance (σ^2) etc are parameters.
- **Statistic**: A quantity computed from the observations in a random sample is called statistic. Mean ($\bar{y}$), variance (s^2) are statistic.

Measures	Parameter	Statistic
Mean	μ	$\bar{y}$
Standard deviation	σ	s
Number	N	n
Proportion	P	p
Correlation	ρ	r

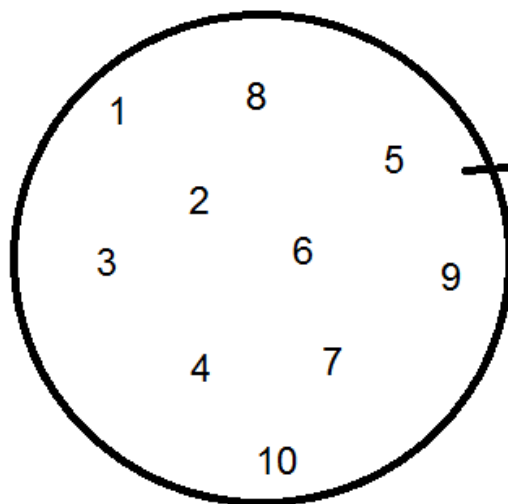
- **Estimator**: Any sample statistic that is used to estimate a population parameter is called an estimator.
- **Estimate**: It is a specific observed value of a statistic. In practice, parameter values are not known and estimates based on sample values are generally used.

Sampling distribution of statistic

- If we draw a sample of size n from finite population N , total number of possible samples are $C(N,n) = k$ (say).
- For each of these k samples, we can compute some statistic Mean ($\bar{y}$), standard deviation (s) or sample proportion (p).
- The set of values of the statistic so obtained is called sampling distribution of statistic.

Population, $N = 10$
Population mean = 5.5

Population S.D, $\sigma = 2.87$



sample, $n=3$

- i) 1, 4 8
- ii) 2, 6, 10
-
-
- ...

$$10C3 = 120$$

<u>Mean</u>	<u>Standard dev.</u>
4.33	2.86
6	3.26
.....	
.....	
y_k	s_k

Standard Error:

- The standard deviation of sampling distribution of a statistic is known as its standard error.
- If a random sample of size n is taken from a population having the mean μ and variance σ^2 , then $\bar{y}$ is a random variable whose distribution has the mean μ .
- For sample from infinite population the standard error of sampling distribution of mean is $\frac{\sigma}{\sqrt{n}}$

Standard Error of means

- If random sample of size n are drawn without replacement from a finite population (when sampling fraction $n/N \geq 0.05$), then S.E. of mean = $\frac{\sigma}{\sqrt{n}} * \sqrt{\frac{N-n}{N-1}}$
- If $n/N < 0.05$, $\sqrt{\frac{N-n}{N-1}}$ need not be used

2.6 The sampling distribution of the sample mean:

- If all possible random sample size n are selected from a population with mean μ and standard deviation σ , the mean of the sample means is denoted by $\mu_{\bar{y}} = \mu$
- Also, the standard deviation of the sample mean is denoted by $\sigma_{\bar{y}} = \sigma/\sqrt{n}$.

- In practice, σ is rarely known, we can approximate σ by the sample estimate,

$$s = \sqrt{\frac{\sum_{i=1}^n (Y_i - \bar{Y})^2}{n - 1}}$$

- For large sample size ($n \geq 30$), it is acceptable to use S in place of σ , under these condition, the standard Normal approximation still works.

$$Z = \frac{\bar{Y} - \mu}{S / \sqrt{n}} \sim N(0, 1)$$

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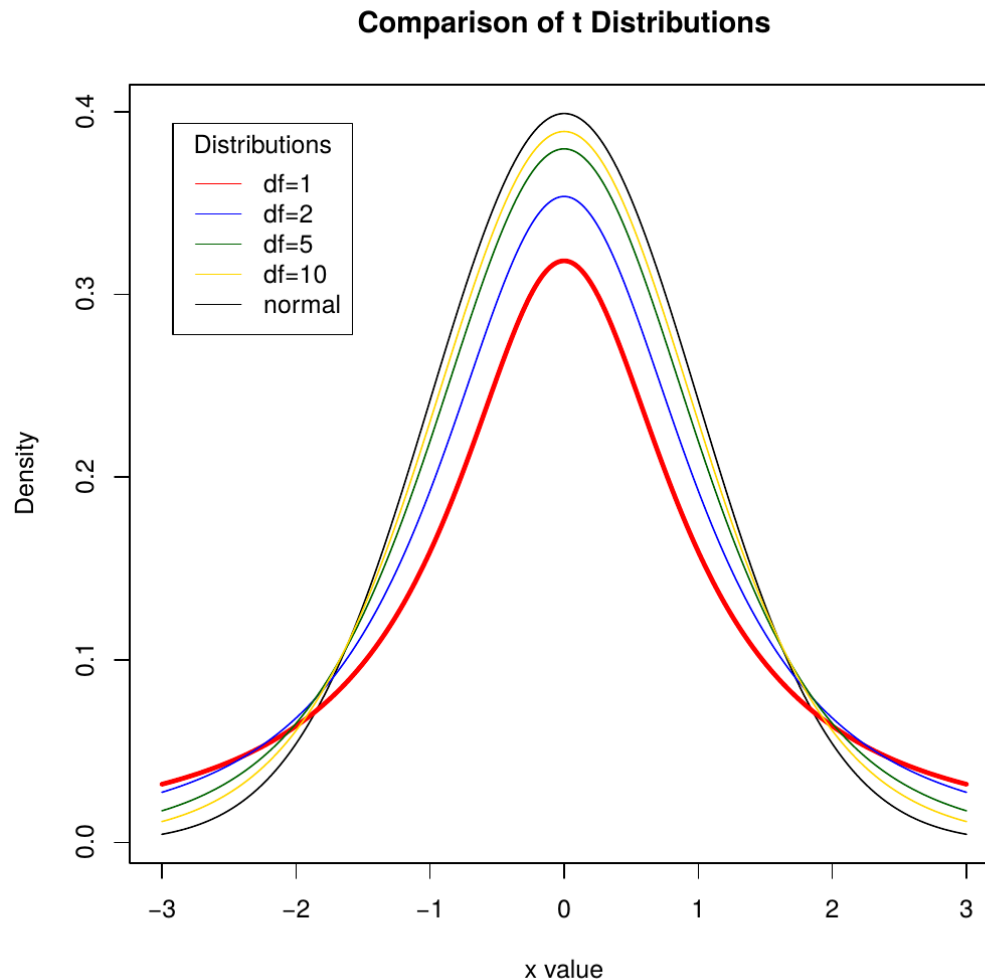
- However, in small samples such as $n < 30$, even if the observations are normally distributed, using above formula does not give good approximation to normal distribution.
- In this case, we use Student's t distribution with $n-1$ degree of freedom.
- Instead of Z , we now use the notation t for above statistic.

$$t = \frac{\bar{Y} - \mu}{s / \sqrt{n}}$$

Properties of t-distribution:

- The shape of the t distribution changes as n increases or decreases
- t is symmetric about zero but is more spread out than the standard normal distribution.
- As the sample size n gets larger, the t-distribution gets closer to the standard normal distribution.

- Different shapes of t-distributions with different degrees of freedom



Assumptions required for t distribution

- The population from which samples are selected at random, is normal
- Population standard deviation is not known.
- Sample size is small ($n < 30$)

Sampling distribution of sampling proportion

- The sampling distribution of a sample proportion is constructed in similar manner as in the case of the mean.
- From the population, which we assume to be finite, we would take all the possible samples of a given size (n) and for each sample we compute the sample proportion, $p = x / n$, where x is no of favorable case in sample.
- We would then prepare a frequency distribution of 'p' by listing the different distinct values of 'p' along with their frequencies of occurrences.
- This frequency distribution (as well as the corresponding relative frequency distribution would constitute sampling distribution of proportion.



Population
proportion p

SRS size n

$\hat{p}$

SRS size n

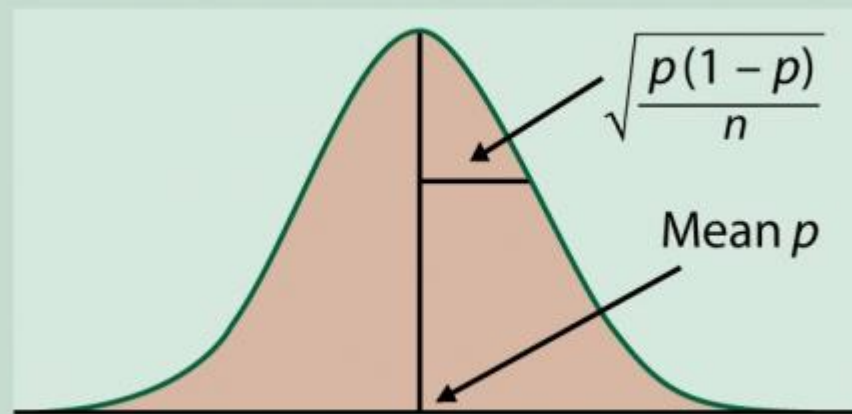
$\hat{p}$

SRS size n

$\hat{p}$

$\vdots$

$\vdots$



← Values of $\hat{p}$ →

- When the sample size is large, the distribution of sample proportion is approximately normally distributed by virtue of the central limit theorem
- The mean of distribution is average of all possible sample proportion, will equal to true population proportion P ,
- Standard error of the distribution $\sigma_p = \sqrt{\frac{PQ}{n}}$
- Thus, $z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} \sim N(0,1)$

2.7 Central Limit Theorem (CLT):

The central Limit theorem states that, under rather general conditions, sums and means of random samples of measurement drawn from the population tend to have an approximately Normal distribution.

Definition:

- If random samples of n observation are drawn from a non-normal population with finite mean μ and standard deviation σ , then, when n is large, the sampling distribution of the sample mean $\bar{y}$ is approximately a normal distribution, with mean μ and standard deviation $\sigma/\sqrt{n}$. The approximation becomes more accurate as n becomes large.
- If $\bar{y}$ is the mean of random sample size n taken from a population having the mean μ and the standard deviation σ , then

$Z = \frac{\bar{y} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$ is a random variable whose distribution function approaches that of the standard normal distribution as $n \rightarrow \infty$.

Central Limit Theorem in terms of Sum

- If $Y_1, Y_2, \dots, Y_n$ be random samples having mean μ and variance σ^2 , then for large n , the distribution of sum $S_n = Y_1 + Y_2 + \dots + Y_n$ is approximately normal with mean $n\mu$ and variance $n\sigma^2$.
- $$Z = \frac{S_n - n\mu}{\sigma\sqrt{n}} \sim N(0,1)$$